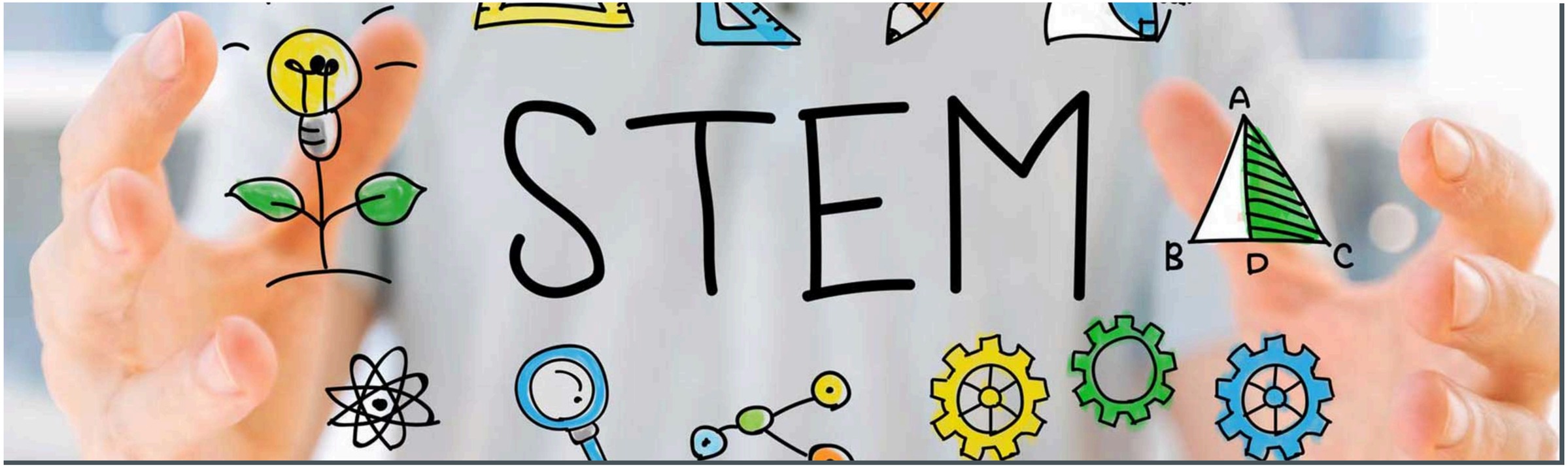
Study Buddy

STEM AI Integrated AI-Powered STEM Learning Ecosystem for Personalized, Adaptive & Multimodal Education

Project ID : R26-IT-009



Members

IT22087638
ELWITIGALA S.A.I.

IT22136756
MANODYA K .M. B.

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COMMENTS FOR THE PROPOSAL PRESENTATION

Topic	Early STEM learning through AR storytelling			
Project ID	R26-IT-009			
Specialization	IT			
Student ID	IT22087638	IT22136756	IT22272768	IT22292568
PanelName	IT-P01			
Detailed Comments	* student has somewhat of a an idea of the task but needs to focus on the actual practicality of the solution * lacks the knowledge of the idea being presented	* lacks knowledge or plan to actually the proposed ideas and concepts	* scope is insufficient for the research based task	* component is too simliar to the topic presented by the first member and lacks scope
Our Improvements	**O/L Focus:** Shifted from Early STEM for impact. **RAG Logic:** Replaced AR for device accessibility. (Practical Issues)	Conducted research and gained deeper knowledge about the concept, while planning proper components with innovative and novel features.	Enhanced the project scope through proper component conceptualization, detailed research, and knowledge gathering to develop innovative and effective solutions.	Redesigned the component with unique concepts and features to differentiate it from the first member's contribution.Increased the scope through additional research, innovative planning, and distinct implementation objectives.

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Real World Problem VS Result

Traditional STEM education in Sri Lankan schools mainly relies on:

- Textbook-based learning
- One-way classroom teaching
- Memorization-focused assessments
- Limited personalized support

As Result, many students experience:

- Low engagement in STEM subjects
- Difficulty understanding abstract scientific concepts
- Lack of adaptive learning support
- Stress during assessments
- Poor knowledge retention
- Limited access to personalized tutoring and revision assistance

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Objective

StudyBuddy STEM AI aims to deliver an AI-powered personalized STEM learning ecosystem that improves:

- Student engagement
- Concept understanding
- Adaptive learning experiences
- Emotional learning support
- Knowledge retention
- Independent STEM learning

through intelligent conversational tutoring, personalized storytelling, adaptive assessments, and AI-driven knowledge synthesis.

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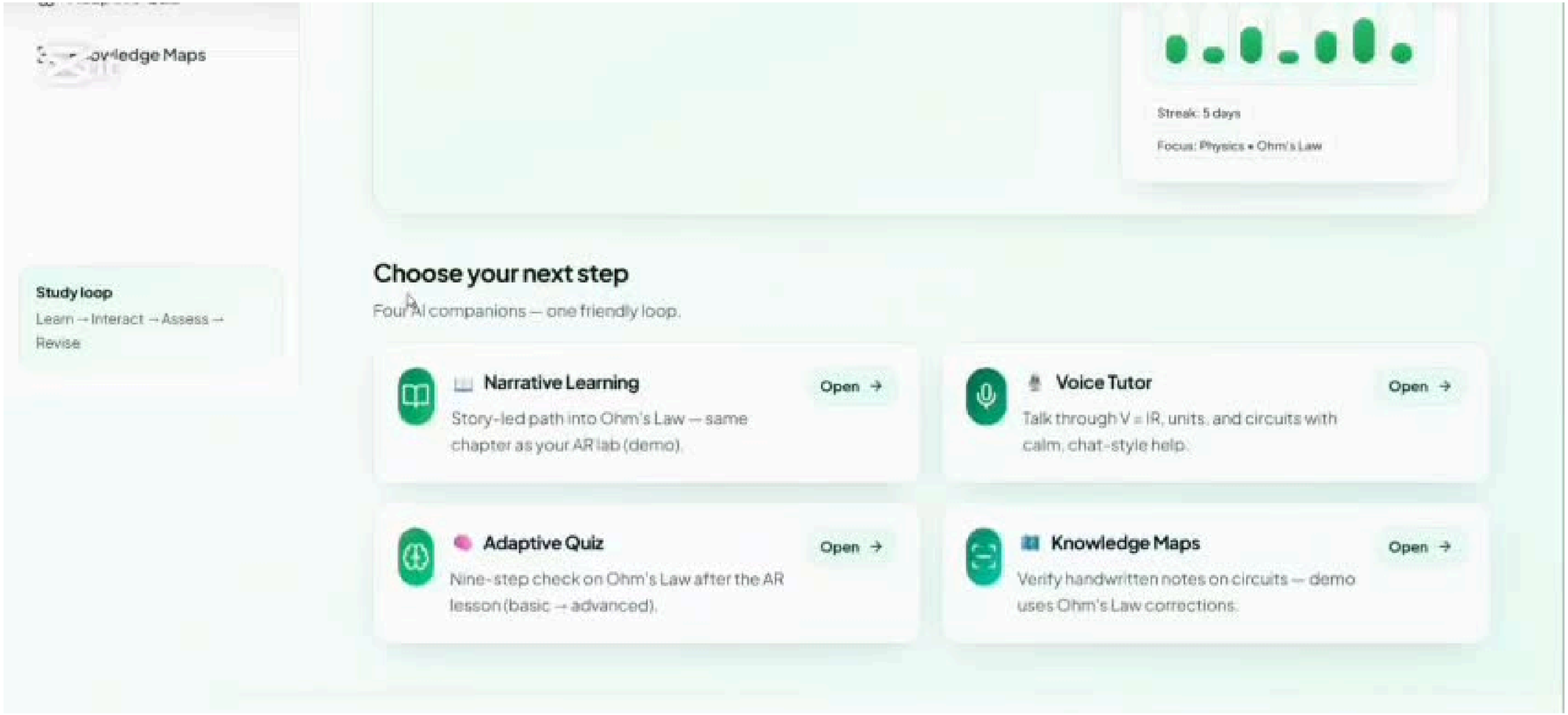
DEMO VIDEO



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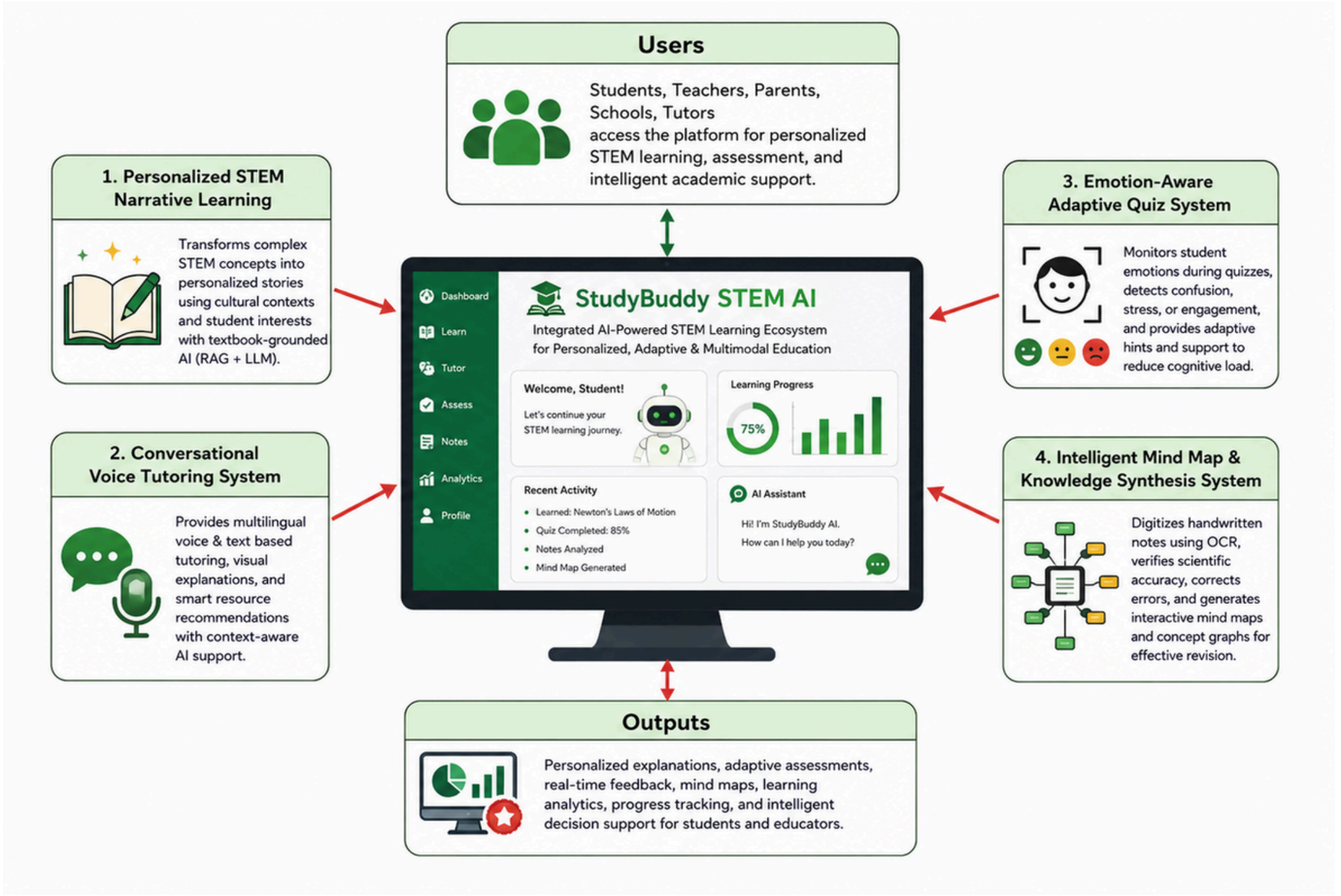
OVERALL SYSTEM DIAGRAM



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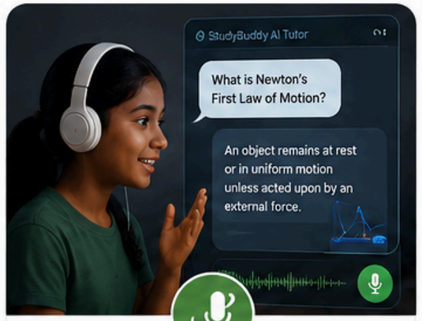
INDIVIDUAL STUDENT COMPONENTS

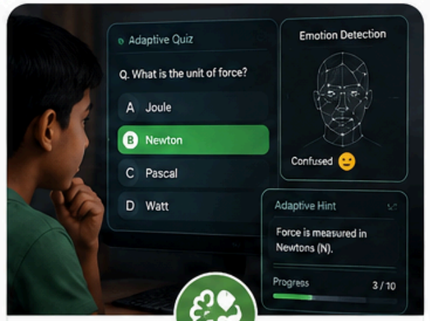
Member 1 – Personalized STEM Narrative Learning

- Transforms complex STEM concepts into personalized stories using cultural contexts and student interests.
- Uses Retrieval-Augmented Generation (RAG) with textbook-grounded AI.
- Provides engaging, context-aware explanations to improve concept understanding.
- Example contexts: Sports, Farming, Transport, Village life, Urban experiences.



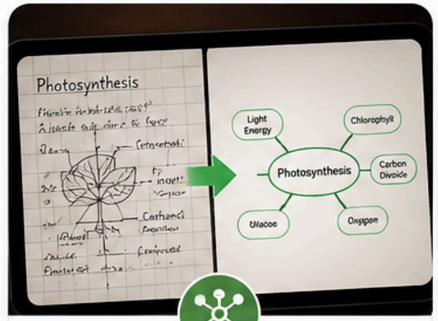
Member 2 – Conversational Voice Tutoring System

- AI-powered multilingual voice and text tutor for STEM subjects.
- Provides contextual explanations, diagrams, and real-life examples.
- Supports voice input/output for natural conversations.
- Recommends videos, notes, and practice questions based on the student's needs.



Member 3 – Emotion-Aware Adaptive Quiz & Assessment System

- Monitors student emotions in real-time during assessments.
- Detects confusion, stress, frustration, or engagement.
- Provides adaptive hints and motivational support.
- Adjusts difficulty and question types based on emotional and performance insights.



Member 4 – Intelligent Mind Map & Knowledge Synthesis System

- Digitizes handwritten notes using OCR.
- Verifies scientific accuracy and detects misconceptions.
- Provides corrections and detailed explanations.
- Generates interactive mind maps and concept graphs for effective revision and better retention.

RAG Based generative and personalized STEM storytelling.

IT22087638

ELWTIGALA S.A.I

BACKGROUND

- Students struggle to bridge the gap between abstract theoretical science and practical real-world application using traditional learning methods.
- Existing educational AI tools provide static, one-dimensional explanations that fail to resonate with a student's unique psychological profile.
- Many system lacks -
  - personalized story contexts
  - interactive scenario applications
  - localized syllabus integration
  - rigorous factual grounding
  - behavioral verification through branching

Proposed Solution

An AI-driven Adaptive STEM Narrative Simulator featuring::

- persona-based context generation
- syllabus-anchored RAG retrieval
- interactive branching narrative logic
- similarity-indexed factual validation

RESEARCH GAP

Research Gap Analysis: STUDY BUDDY Component

	Stem Narrative Pedagogy	Adaptive Learner Profiling (Classifier)	Factual Branching Logic	Localized Syllabus Anchor (NIE/OL)
Traditional Rote-Learning (Textbooks) Static, non-adaptive.	✗	✗	✗	✓
Generic Generative AI (Chat/Story) Creative, non-anchored.	✓	⚠	✗	✗
Static RAG Systems (Fact-Retriever) Linear, passive fact delivery.	✗	✗	✗	✓
STUDY BUDDY Personalized Narrative Simulation. (G.C.E. O/L STEM)	✓	✓	✓	✓

(G.C.E. O/L English Medium STEM Science)

Gap Identified :

Existing Systems fail to combine

- narrative-based pedagogy
- syllabus-anchored retrieval
- interactive branching logic
- adaptive persona modeling

-into one unified storytelling framework.

RESEARCH OBJECTIVE

Research Objective

Develop an **Adaptive Narrative Pedagogy System (ANPS)** component for Study Buddy that delivers personalized STEM learning through:

- **Personalized Narrative Generation**
- **Predictive Learner Classification**
- **Story-Based Visual Synthesis**
- **Factual Branching & Verification**

Research Sub-Objectives

- Refine Multilingual Story Delivery
- Implement Syllabus-Anchored RAG Retrieval
- Track and Adapt to Learner Decision Paths
- Synthesize Context-Aware Narrative Visuals
- Deliver Adaptive, Consequence-Based Lessons
- Synthesize Text Narratives with Visual Branching

FUNCTIONAL & NON – FUNCTIONAL REQUIREMENTS

Functional Requirements

1. Personalized Narrative Pedagogy
Adapts story themes to student interests and career goals.
2. Syllabus-Anchored Retrieval
Anchors all story content to official NIE textbooks.
Mathematical Similarity Scoring Verifies scientific accuracy of generated narratives.
3. Interactive Plot Generation Generates:

- adaptive story chapters
- diagnostic decision points
- consequence-based scenarios

4. Behavioral Verification Validates:

- conceptual application
- misconception patterns

Non-Functional Requirements

- Performance
- Accuracy
- Reliability
- Scalability
- Usability
- Accessibility
- Cross-device support

TECHNOLOGIES

Frontend

- **Streamlit** — Interactive story flow.
- Adaptable persona interface.

Backend

- **Python** — RAG orchestration.
- Diagnostic branching logic.

Database

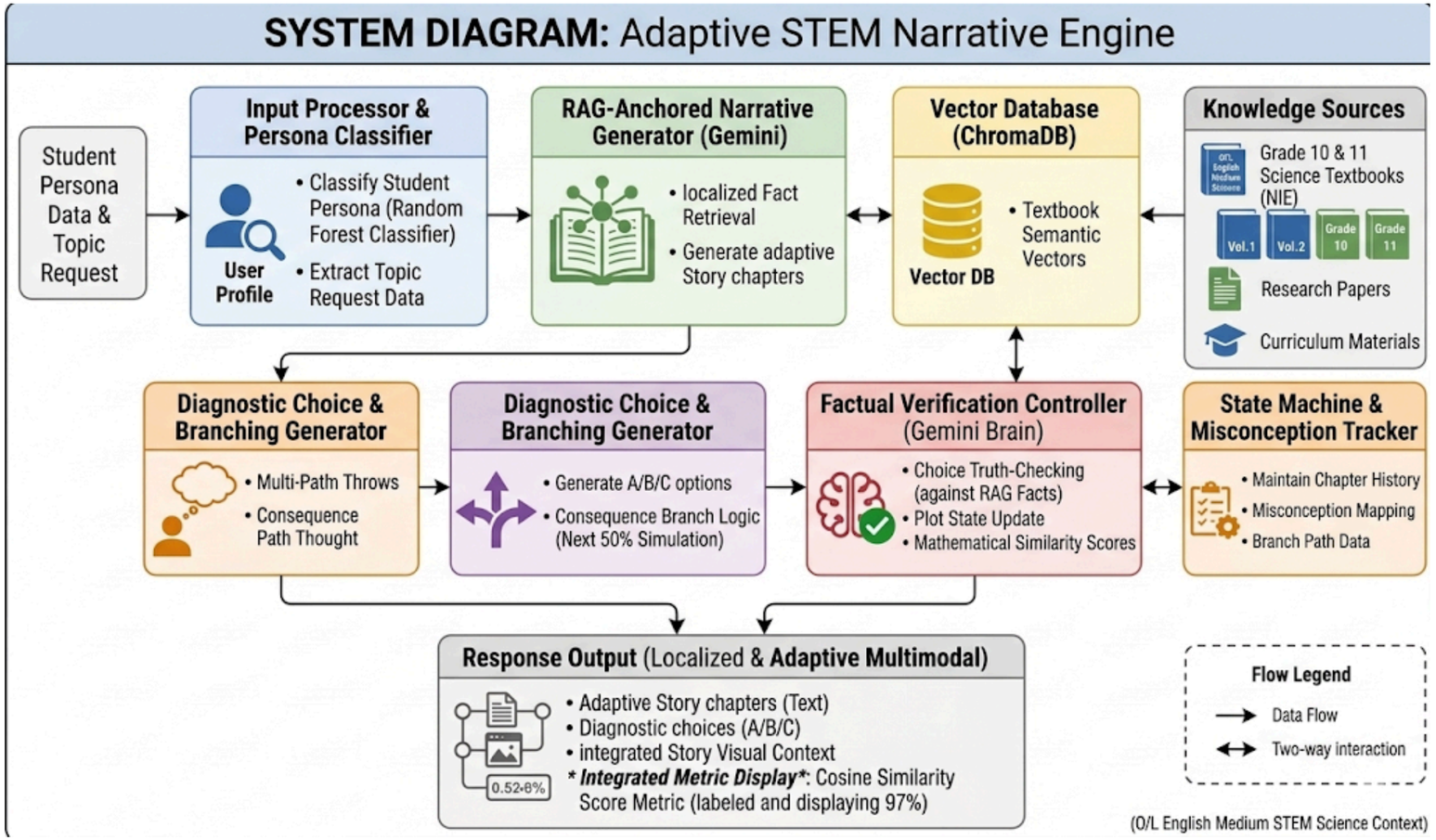
- **ChromaDB** — NIE Textbook vectors. Semantic concept storage.
- **MongoDB** — Document store. Student profile & quiz response storage.

ML & AI

- **RAG** (LangChain) — Fact retrieval.
- **Random Forest** — Theme mapping.
- **scikit-learn** — Classifier metrics.

Other: ChromaDB (Vector store), **MongoDB** (Document store), **LangChain** (Orchestration), Google GenAI & Embeddings), **PyMuPDF** (Syllabus text extraction).

SYSTEM DIAGRAM



COMPLETION OF COMPONENT

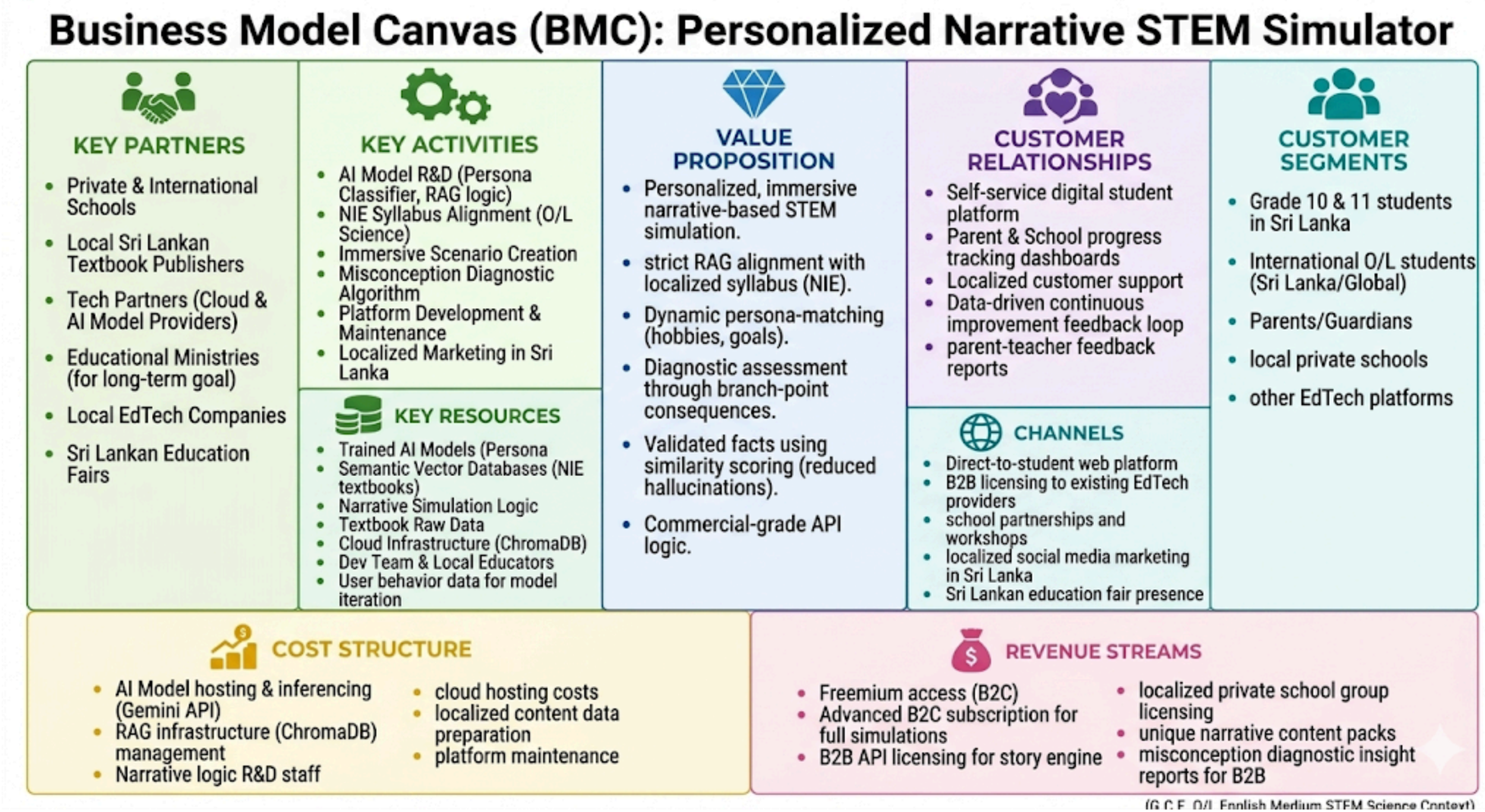
Completion up to Progress Presentation 01

- Initial RAG Pipeline Development
- Syllabus Dataset (NIE) Extraction & Preprocessing
- Syllabus-Anchored Narrative Generation Prototype
- Semantic Vector Database Implementation (ChromaDB)
- Metadata Mapping for Textbook Citations
- Initial Narrative Chapter State Management
- Streamlit-based prototype Story Interface Development
- API Integration for LLM Narrative Flow

Completion up to Progress Presentation 02

- Implement ML Persona Classification (Random Forest)
- Integrate Dynamic Narrative-to-Infographic Synthesis
- Enhance Interactive Branching Decision Logic
- Implement Diagnostic Misconception Mapping System
- Integrate Mathematical Similarity Score Metrics
- Optimize Consequence-based Plot Generation
- Complete Adaptive Persona–Narrative Integration

Ability of Commercialization



MULTILINGUAL KNOWLEDGE RETRIEVAL & CONVERSATIONAL AI SYSTEM

IT22136756

Manodya K.M.B.

BACKGROUND

- Students struggle to understand complex STEM concepts using traditional learning methods.
- Existing educational chatbots provide generic answers without contextual understanding.
- Many systems lack:
  - personalized tutoring
  - visual explanations
  - multilingual interaction
  - factual grounding
- AI hallucinations can produce inaccurate educational responses.

Proposed Solution

An AI-powered conversational STEM tutoring system with:

- contextual understanding
- RAG-based factual retrieval
- visual explanation generation
- personalized learning support

RESEARCH GAP

System / Research	Context Awareness	Visual Support	Personalized Recommendation	Hallucination Reduction
Traditional Chatbots	✗	✗	✗	✗
Standard RAG Systems	⚠	✗	✗	✓
Voice Tutoring Systems	✗	✗	✗	✗
StudyBuddy STEM AI	✓	✓	✓	✓

Gap Identified

Existing systems fail to combine:

- contextual memory
- multimodal tutoring
- visual learning support
- intelligent recommendations

into one unified educational platform.

RESEARCH OBJECTIVE

Research Objective

Develop a context-aware multimodal AI tutoring system that provides personalized STEM learning assistance through:

- Conversational AI
- Contextual Understanding
- Visual Explanations
- Intelligent Educational Recommendations

Research Sub-Objectives

- Develop multilingual voice & text tutoring support
- Implement RAG-based factual retrieval
- Develop contextual conversation tracking
- Generate AI-powered visual explanations
- Recommend personalized educational resources
- Support multimodal interaction:
  - voice
  - text
  - visuals
  - audio narration

FUNCTIONAL & NON – FUNCTIONAL REQUIREMENTS

Functional Requirements

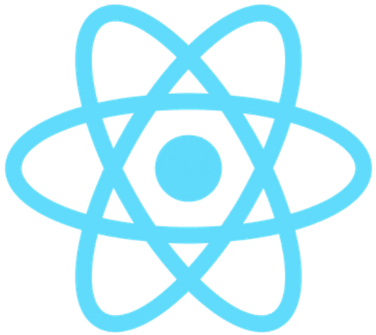
1. Voice & Text Interaction
Supports student questioning using voice and text.
2. Context Tracking
Maintains previous learning conversations.
RAG-Based Retrieval
Retrieves verified STEM knowledge.
3. Visual Explanation Generation
Generates:

- diagrams
- flowcharts
  concept illustrations
  4. Personalized Recommendations
  Suggests:
- videos
- Simulations

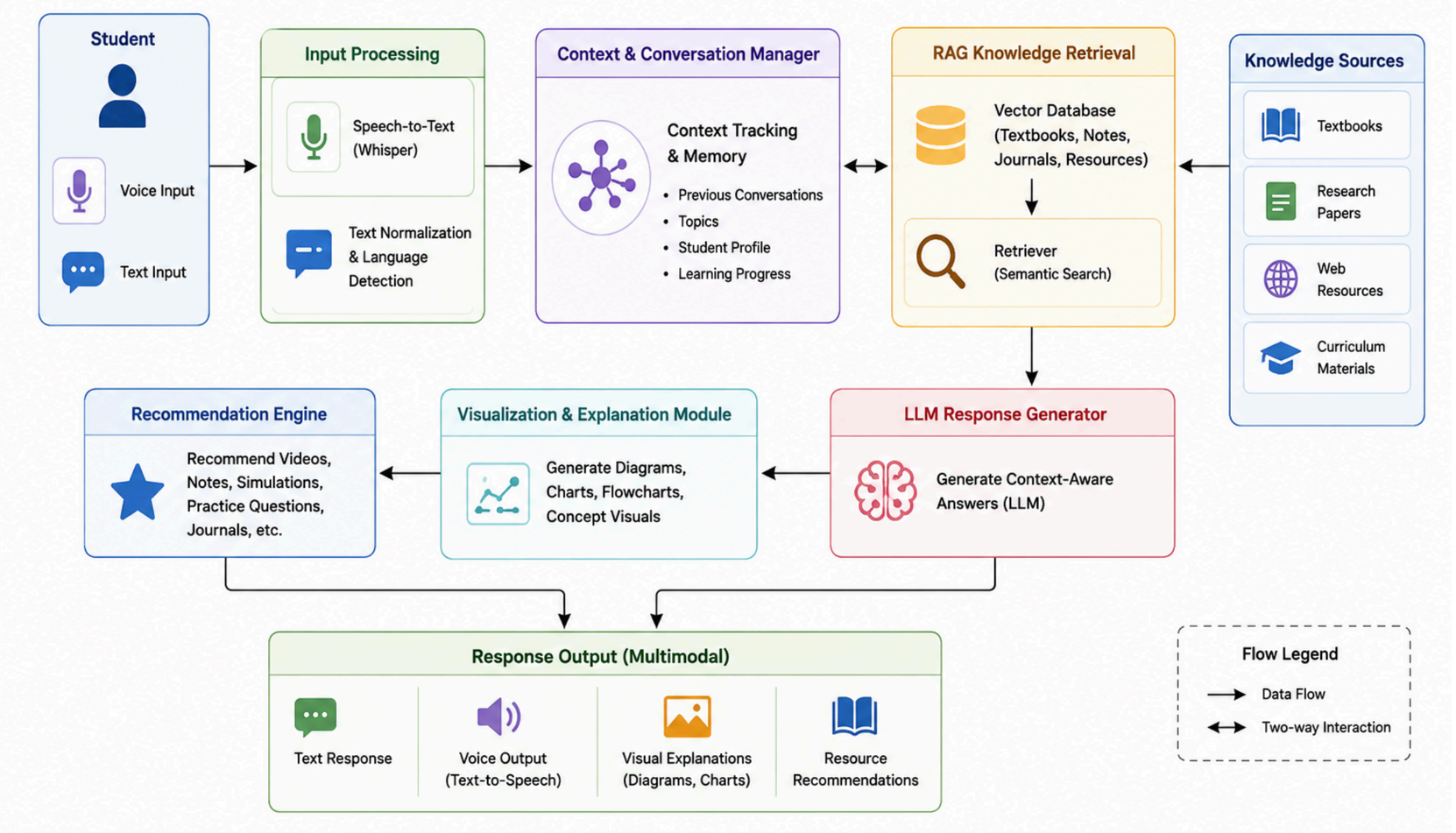
Non-Functional Requirements

- Performance
- Accuracy
- Reliability
- Scalability
- Usability
- Accessibility
- Cross-device support

TECHNOLOGIES



SYSTEM DIAGRAM



COMPLETION OF COMPONENT

Completion up to Progress Presentation 01

- Initial RAG Pipeline Development
- Educational Dataset Collection & Preprocessing
- Conversational AI Prototype Development
- Speech-to-Text Integration using Whisper
- Context Tracking & Conversation Memory Implementation
- Frontend Chat Interface Development
- Initial Recommendation Engine Prototype
- API Integration for LLM Communication

Completion up to Progress Presentation 02

- Improve Context-Aware Conversational Memory
- Integrate Dynamic Visual Explanation Generation
- Enhance Multilingual Tutoring Support
- Implement Personalized Learning Recommendation System
- Optimize Real-Time AI Response Performance
- Integrate Voice Narration (Text-to-Speech)
- Complete Full Frontend–Backend System Integration

Emotion-Aware Adaptive Quiz & Assessment System

IT22272768

BASNAYAKA W.B. M.D.K.

BACKGROUND

- Students struggle to learn STEM concepts using traditional assessment methods. Existing quiz systems treat all students the same without adapting to their needs.

Many systems lack:

  - Personalized difficulty adjustment
  - Real-time emotion detection
  - Behavioral monitoring
  - Adaptive feedback
  - Cross-session learning tracking

Fixed assessments cannot identify individual student weaknesses effectively.

Proposed Solution

An AI-powered behavior-aware adaptive assessment system with:

- Real-time facial expression detection
- ML-based learning state prediction
- Adaptive difficulty adjustment
- Personalized hint and feedback delivery
- Cross-session weak area prioritization

RESEARCH GAP

System / Research	Emotion Detection	Behavioral Monitoring	Adaptive Difficulty	Personalized Feedback
Traditional Quiz Systems	✗	✗	✗	✗
Static Adaptive Systems	✗	✗	⚠	✗
Rule-Based Tutoring Systems	✗	✗	✗	⚠
Your System	✓	✓	✓	✓

Gap Identified

Existing systems fail to combine:

- Real-time emotion detection
- Behavioral signal analysis
- ML-based learning state prediction
- Adaptive difficulty adjustment
- Personalized feedback delivery

RESEARCH OBJECTIVE

Research Objective

Develop a behavior-aware adaptive assessment system that provides personalized STEM quiz experiences through:

- Real-Time Emotion Detection
- ML-Based Learning State Prediction
- Adaptive Difficulty Adjustment
- Personalized Feedback Delivery

Research Sub-Objectives

- Detect student facial expressions in real-time using YOLO model every 1.5 seconds
- Predict learning state using Random Forest classifier from 5 behavioral features
- Adjust question difficulty dynamically based on cognitive load score
- Deliver personalized hints and theory explanations based on predicted learning state
- Prioritize weak concept areas from previous sessions in next assessment attempt

FUNCTIONAL & NON – FUNCTIONAL REQUIREMENTS

Functional Requirements

1. Behavioral Data Collection Capture response time, answer changes and facial expression per question.
2. Emotion Detection Detect facial expressions every 1.5 seconds using YOLO model.
3. Learning State Prediction Predict student understanding level using Random Forest model.
4. Adaptive Feedback Show hint or theory explanation based on prediction.
5. Adaptive Difficulty Adjust question level based on student performance and cognitive load.
6. Report Generation Show score, weak areas and study plan after assessment.

Non-Functional Requirements

- Performance — Emotion detection within 2 seconds
- Accuracy — Noise filtering using confidence threshold and majority vote
- Reliability — Works even if ML service is down using fallback logic
- Scalability — Cloud database supports multiple students
- Security — JWT tokens and encrypted passwords
- Usability — Accessible on any modern browser

TECHNOLOGIES

Frontend

- User interface and quiz flow

Backend

- FastAPI server, authentication, database logic

Database

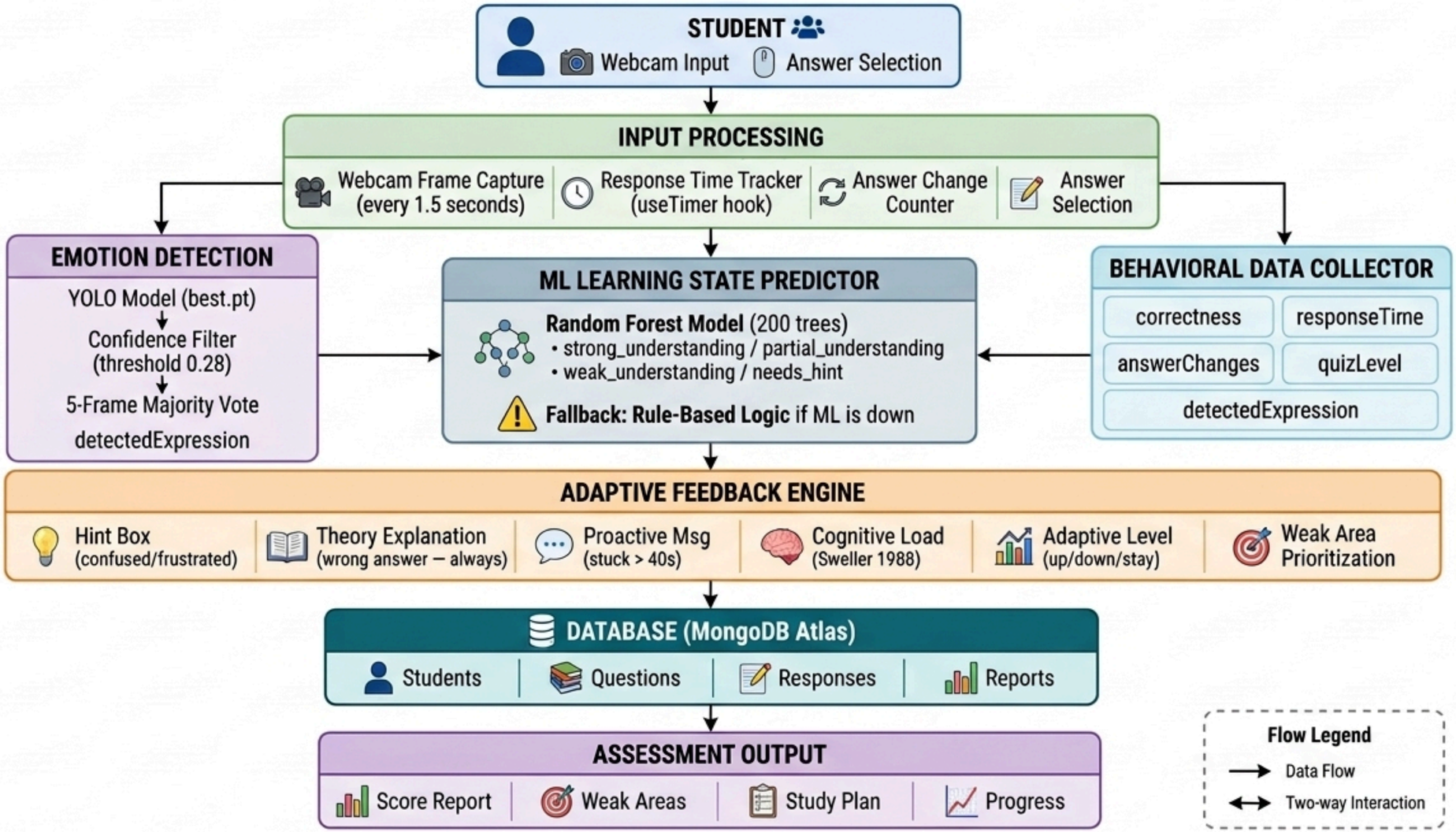
- Stores students, questions, responses and reports

ML & AI

- **YOLO** — Real-time facial expression detection from webcam
- **Random Forest** — Learning state prediction
- **T5 Model** — Question generation from PDF
- **scikit-learn** — ML model training and evaluation

Other: JWT + **bcrypt** (Secure authentication), **Web Speech API** (Voice command support), **Ultralytics** (**YOLO** model inference), **HuggingFace** (T5 model for question generation)

SYSTEM DIAGRAM



COMPLETION OF COMPONENT

Completion up to Progress Presentation 01

- Database Design and MongoDB Atlas Setup
- Basic Quiz Flow Development (3 levels)
- Theory Explanation and Hint Delivery System
- Webcam Integration and Real-Time Emotion Detection
- Model Integration for Facial Expression Analysis
- Per-Question Timer Implementation
- Basic Assessment Report Generation

Completion up to Progress Presentation 02

- Full Assessment Report Generation
- Cross-Session Weak Area Prioritization
- Personalized study plan generation based on weak areas
- 5-frame majority vote smoothing for expression noise reduction
- Model Integration for Automatic Quiz Question Generation
- Complete Full Frontend–Backend System Integration

Handwritten Content Analysis, Intelligent Mind Map

&

Knowledge Synthesis System

IT22292568
Dilshan WA

BACKGROUND

- Students struggle to manually synthesize complex STEM text into structured study aids (Mind Maps, Flashcards).
- Existing tools cannot generate personalized study aids (flashcards, mind maps, structured notes) from raw text.
- OCR tools fail to extract Sinhala or mixed-language handwritten content accurately.
- AI-generated educational content often contains hallucinations with no factual grounding.
- No automated way to check student notes against official Sri Lanka MOE syllabus content.

Proposed Solution

An AI-powered conversational STEM tutoring system with:

- Automated synthesis (Mind maps, notes)
- RAG-based factual retrieval
- visual explanation generation
- personalized learning support

RESEARCH GAP

System / Research	Knowledge Synthesis	Syllabus Grounding	Handwriting OCR	Web Fallback
Traditional Chatbots	✗	✗	✗	✗
Standard RAG Systems	!	✓	✗	✗
Flashcard Apps	✓	✗	!	✗
StudyBuddy (M4)	✓	✓	✓	✓

Gap Identified

Existing systems fail to combine:

- structured synthesis (mind maps, structured notes, flashcards)
- visual OCR processing for handwritten student notes
- automated factual claim verification grounded strictly in local syllabuses

RESEARCH OBJECTIVE

Research Objective

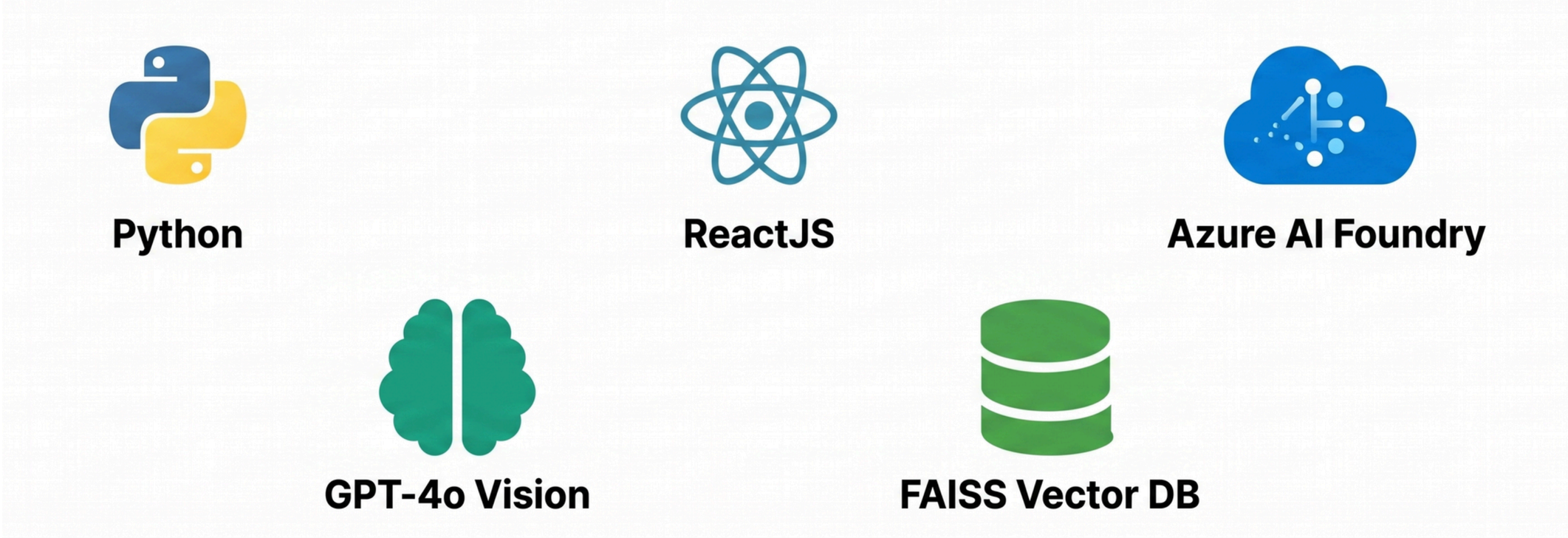
Develop an intelligent knowledge synthesis and verification engine using Azure AI that provides structured study aids and multimodal fact-checking through:

- **Knowledge Synthesis Engine**
- **Syllabus Claim Verification**
- **Visual Handwritten OCR**
- **Web Evidence Fallback**

Research Sub-Objectives

- Implement Azure GPT-4o tool-calling to generate Mind Maps, Flashcards, and Notes.
- Develop a 3-stage verification pipeline (Extraction → Syllabus Grounding → Web Evidence).
- Deploy GPT-4o Vision for high-fidelity OCR extraction from handwritten science notes.
- Utilize FAISS index to strictly ground verifications against Grade 10-11 Syllabuses.

TECHNOLOGIES



FUNCTIONAL & NON – FUNCTIONAL REQUIREMENTS

Functional Requirements

1. Knowledge Synthesis Generation
Generates structured Mind maps, Flashcards, and Notes from text inputs using tool-calling.
2. Context Tracking
Maintains previous learning conversations.
RAG-Based Retrieval
Retrieves verified STEM knowledge.
3. Visual Handwritten OCR
Extracts science formulas and text from uploaded images using GPT-4o Vision.

Non-Functional Requirements

1. Performance (Synthesis < 5 seconds)
2. Accuracy (Strict syllabus grounding)
3. Reliability (Web fallback integration)
4. Scalability (Concurrent image processing)
5. Usability (JSON output for UI)
6. Accessibility (Bilingual output)

COMPLETION OF COMPONENT

Completion up to Progress Presentation 01

- Azure AI Foundry Environment Setup
- OCR Module via GPT-4o Vision (image → text extraction)
- Deploymentof Singleton Azure OpenAI Client
- FAISS index setup
- Basic API routing development
- Initial architecture design for verification
- Auto Topic Tag Extraction from Corrected Notes
- Initial Architecture Design for Claim Verification

Completion up to Progress Presentation 02

- Improve Knowledge Synthesis (Mind Map & Flashcard part)
- Complete Full Frontend–Backend System Integration (Verify Tab)
- YouTube Video Recommendation & Transcript Summarization
- End-to-End Workflow: OCR → Verify → Correct → Tag → Video → Add to Note

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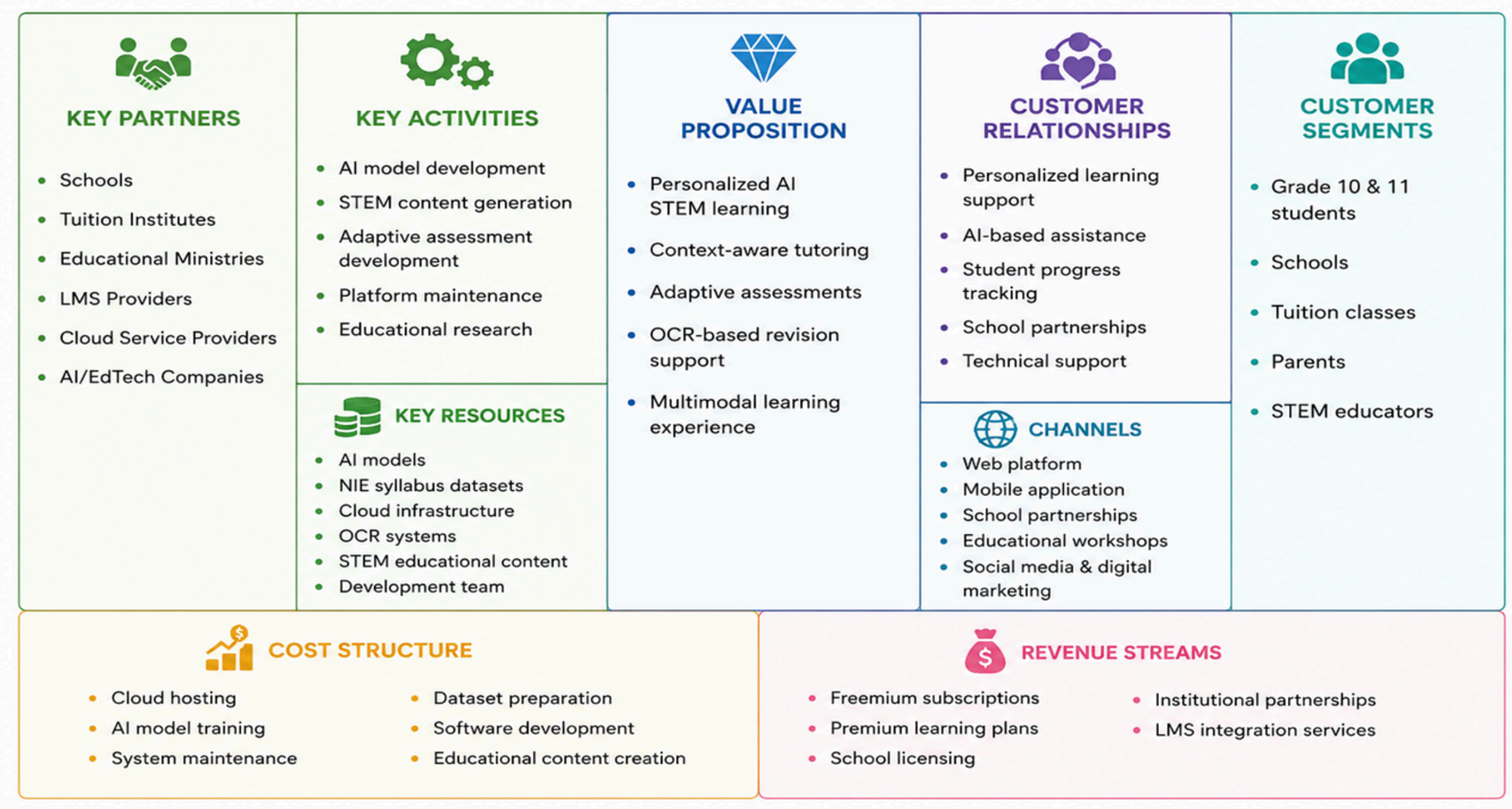
ABILITY OF COMMERCIALIZATION



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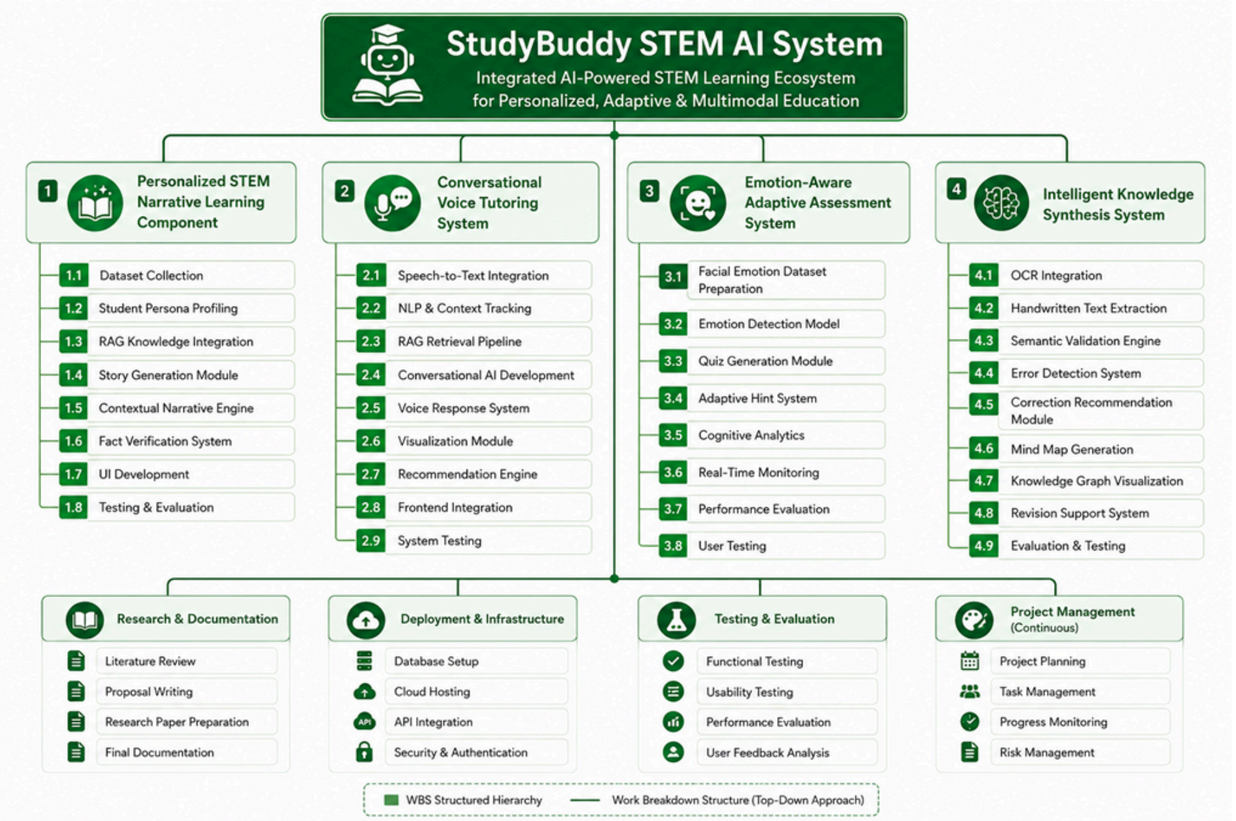
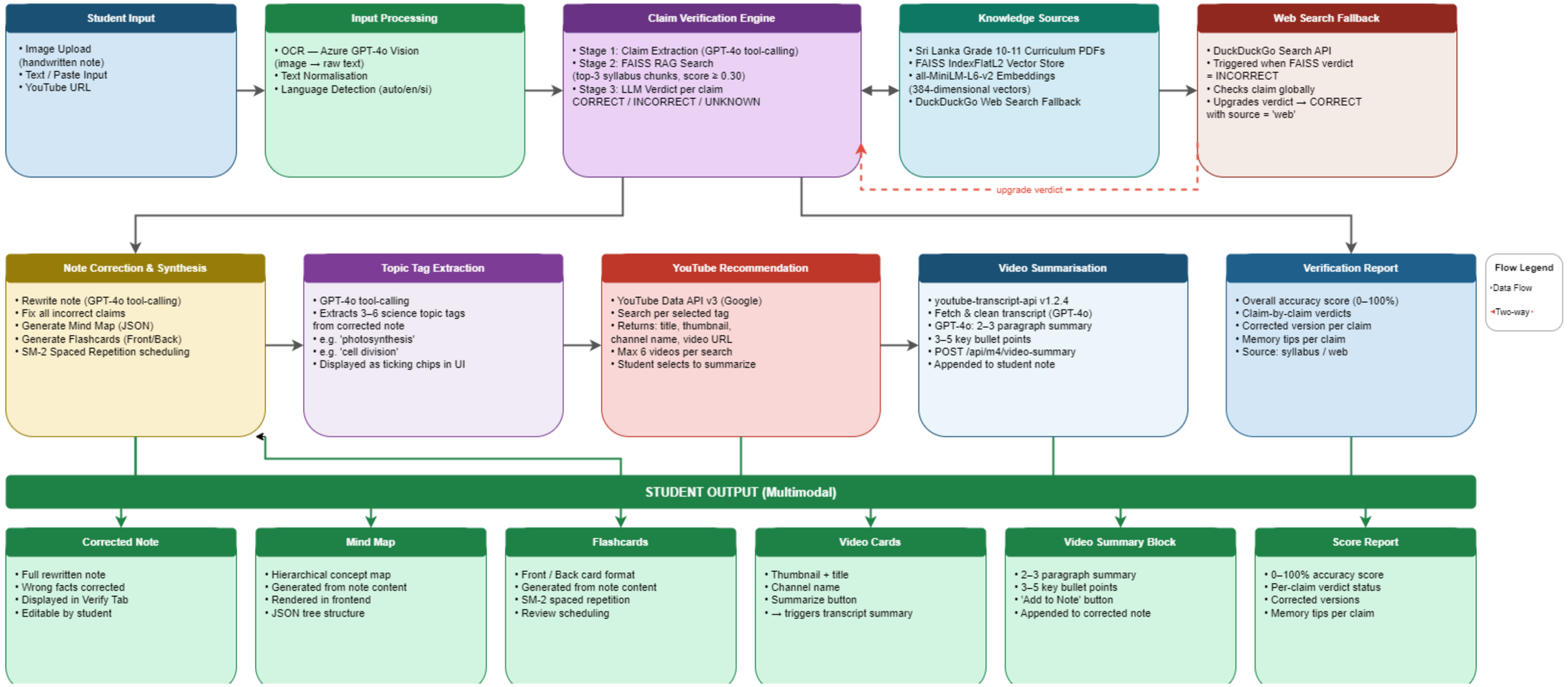
WORK BREAKDOWN STRUCTURE (WBS)



SYSTEM DIAGRAM



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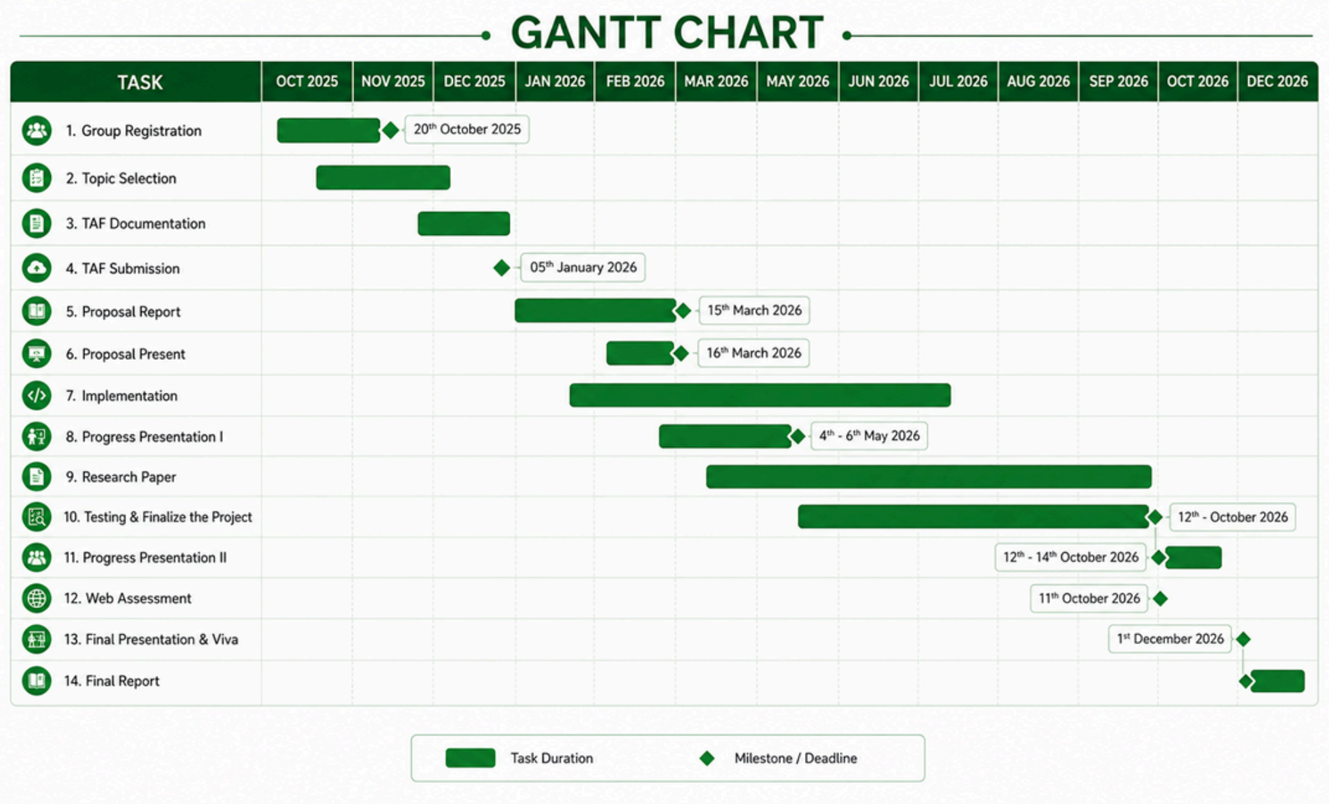
GANTT CHART



THANK YOU…